

Sharper Zarankiewicz and Diagonal Bipartite Ramsey Bounds

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September 26, 2026

Abstract

We prove that there is an absolute positive constant c such that every bipartite graph with N vertices in each part and at least $N^2/2$ edges contains a complete bipartite graph $K_{t,t}$ whenever $N \geq c2^t$. This improves the classical Kővári-Sós-Turán bound requiring N of order $t2^t$. As a consequence, the diagonal bipartite Ramsey number has upper bound $b(t, t) = O(2^t)$, improving the previous best bound $b(t, t) = O(2^t \log t)$ due to Conlon. The proof was found by GPT-6 Astra, and the method will probably have further applications.

1 AI Declaration Statement

The proof of the main result was found by GPT-6 Astra after several guided prompts by the author. Astra first produced a proof that was only 8 pages long and was incomprehensible to the author. As the author tried to understand the proof, many iterations of the write-up were produced, culminating in the current one in which much of the text and explanations are due to the author. The author takes full responsibility for all the content in the paper.

2 Introduction

For positive integers m, n, s, t , let $z(m, n; s, t)$ denote the maximum number of edges in a bipartite graph with specified classes of sizes m, n that contains no complete bipartite graph with s vertices in the first class and t in the second. When $s = t$ we refer to this as the diagonal case. The theorem of Kővári, Sós, and Turán [8] gives, in the diagonal case,

$$z(N, N; t, t) \leq (t-1)^{1/t} N^{2-1/t} + (t-1)N \quad (t \geq 2). \quad (2.1)$$

In a bipartite graph with density at least $1/2$, the counting argument showing (2.1) guarantees a $K_{t,t}$ when N is of order $t2^t$. Our result removes the factor of order t in this dense regime.

Theorem 2.1. *Let t, N be positive integers. If $G \subseteq K_{N,N}$, $e(G) \geq N^2/2$, and $N \geq 2^{t+81}$, then G contains $K_{t,t}$.*

Equivalently, Theorem 2.1 asserts that $z(N, N; t, t) < N^2/2$ for $N \geq c2^t$ where $c = 2^{81}$, whereas substituting $N = c2^t$ into (2.1) gives, for fixed $c > 0$ as $t \rightarrow \infty$,

$$z(N, N; t, t) \leq \left(\frac{1}{2} + \frac{\log t - \log c}{2t} + O_c \left(\frac{(\log t)^2}{t^2} \right) \right) N^2.$$

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This shows explicitly why the Kővári–Sós–Turán estimate alone does not establish the desired conclusion when $N = c2^t$ with c fixed. The exact bound (2.1) requires c to grow on the order of t to be effective in the range $e(G) > N^2/2$. The constant 2^{81} has not been optimized. More generally, Corollary 7.1 shows that a bipartite graph of density $0 < \rho < 1$ with N vertices in each class contains $K_{t,t}$ whenever

$$N \geq C_\rho \rho^{-t}, \quad C_\rho = 2^{41}[\rho(1-\rho)]^{-20}.$$

Thus, for each fixed density, the sufficient number of vertices is a constant multiple of ρ^{-t} ; taking $\rho = 1/2$ recovers the bound $N \geq 2^{t+81}$.

For $1 \leq s \leq t$, let $b(s, t)$ be the least N such that every red–blue coloring of $K_{N,N}$ contains a monochromatic $K_{s,t}$, where the part of size s (and t) can lie in either part of the $K_{N,N}$. The bipartite Ramsey problem was introduced by Beineke and Schwenk [1]; its connection with Zarankiewicz numbers was developed by Irving [7]. Thomason [11] proved

$$b(s, t) \leq 2^s(t-1) + 1. \quad (2.2)$$

In particular, $b(t, t) \leq (t-1)2^t + 1$. Conlon [3] subsequently improved the bound in the diagonal case to

$$b(t, t) \leq (2 + o(1))2^t \log_2 t. \quad (2.3)$$

For lower bounds, independently coloring each edge red or blue with equal probability gives an expected number of monochromatic copies of $K_{t,t}$ equal to $2 \binom{N}{t}^2 2^{-t^2}$. The first-moment argument and Stirling’s formula therefore give $b(t, t) \geq (1/e + o(1))t 2^{t/2}$. Hattingh and Henning [6] used the Lovász Local Lemma to obtain the stronger bound $b(t, t) \geq (\sqrt{2}/e + o(1))t 2^{t/2}$, which remains the best known asymptotic lower bound; see also Conlon [3]. Thus the known lower and upper bounds remain exponentially apart. By choosing a color with at least half the edges in a 2-coloring of $E(K_{N,N})$ and applying Theorem 2.1 we immediately obtain the following improvement over (2.3).

Corollary 2.2. *For every positive integer t , we have $b(t, t) \leq 2^{t+81}$.*

Theorem 2.1 does not improve the exponent $2 - 1/t$ in the classical Zarankiewicz problem with t fixed and N tending to infinity. For a broader account of bipartite extremal problems, see Füredi and Simonovits [4].

Section 3 explains the ideas of the proof and provides extensive motivation for introducing a definition (called the potential function) that is the crucial new ingredient in the proof; we expect that this idea may have further applications. The main technical statement is the one-step density theorem, Theorem 3.1, stated in Section 3.8. Sections 4–6 establish the estimates needed for its proof, which is given in Section 7.1. We then deduce a general density criterion in Section 7.2 and prove Theorem 2.1 and Corollary 2.2 in Section 7.3.

3 Proof ideas

The proof builds a copy of $K_{t,t}$ by choosing edges one by one, unless a preliminary density criterion already supplies the remaining biclique. At each selection step, we restrict our attention to the bipartite graph induced by the neighborhoods of the chosen endpoints, after deleting those endpoints, since all future choices must lie in this residual graph. The main idea is to define a potential function that measures how long we can continue this process: the potential starts as a large number and decreases by only a small amount at each selection step. We first explain this algorithm with more precise notation and the requirements on the potential, then motivate its particular choice (which may not be unique).

3.1 Choosing vertices on both sides

The most straightforward approach is to choose all t vertices on one side and look for t common neighbors on the other. If L, R are the classes of a bipartite graph with $|L| = |R| = N \geq t$ and $N_G(S) = \bigcap_{x \in S} N_G(x)$, then double counting gives

$$\frac{1}{\binom{N}{t}} \sum_{\substack{S \subseteq L \\ |S|=t}} |N_G(S)| = \frac{1}{\binom{N}{t}} \sum_{y \in R} \binom{d(y)}{t}. \quad (3.1)$$

For example, if every vertex in R has degree $N/2$, this average is

$$N \frac{\binom{N/2}{t}}{\binom{N}{t}} = (1 + o(1)) N 2^{-t} \quad \text{when } t^2/N \rightarrow 0.$$

When $N = c2^t$ with c fixed, the average is approximately c . Hence to be guaranteed t common neighbors we would need a factor of order t in the classical counting argument.

Instead, select one vertex on each side at a time. Suppose that i vertices have already been chosen in each class and form a $K_{i,i}$. Keep unused left and right candidate sets, each vertex of which is adjacent to every previously selected vertex on the opposite side. Let J be the graph between these candidate sets. For an edge $xy \in E(J)$, select x, y and replace the candidate sets by

$$L' = N_J(y) \setminus \{x\}, \quad R' = N_J(x) \setminus \{y\}. \quad (3.2)$$

Write $J'_{xy} = J[L', R']$ for the resulting graph. Every $K_{k-1, k-1}$ in J'_{xy} extends to a $K_{k, k}$ in J by adding x, y . Thus it suffices either to find the required biclique directly in the current graph or to choose an edge whose residual graph contains the smaller biclique. The problem in the second case is to leave enough useful candidates to continue.

An ideal calculation explains the exponential growth 2^t as follows. Suppose both candidate sets have size M_i , their density remains $1/2$, and the chosen vertices have half the opposite candidate set as neighbors. Then

$$M_{i+1} = M_i/2 - 1, \quad M_i = \frac{N+2}{2^i} - 2.$$

If $N = c2^t$, then $M_t = c - 2 + 2^{1-t}$, so any constant $c > 2$ leaves a nonempty candidate set after t ideal steps. In an arbitrary graph, the candidate sizes and their density vary together. We therefore attach a nonnegative real number to the current graph and to the number k of vertices still required on each side. This number measures the remaining supply of candidates after accounting for future neighborhood restrictions. We call it a *potential function*. The proof chooses an edge for which this potential falls by only a controlled amount; this one-step inequality replaces the recurrence for M_i .

3.2 Requirements for a potential function

Our next goal is to identify the properties that the potential must satisfy. Before choosing a formula, consider nonnegative potential functions $\mathcal{P}_k(J)$, where J is the current bipartite graph and k is the number of vertices still needed on each side. We want thresholds F_k such that

$$\mathcal{P}_k(J) > F_k \implies J \text{ contains } K_{k,k}. \quad (3.3)$$

We call F_k an *inductive potential threshold*. There are four requirements. First, the potential must be defined for every graph produced by the construction, including graphs with unequal or empty

classes and all densities. Second, positive potential at $k = 1$ must guarantee an edge. We arrange this by assigning potential zero to every edgeless graph, including graphs with an empty class.

Third, for $k \geq 2$, a graph with potential above its threshold must allow the algorithm to proceed. We first apply a preliminary density criterion, proved in [Lemma 5.2](#), to the current graph and to its residual graphs. If one of these tests succeeds, it gives the required biclique directly, using the extension property above. If none succeeds, we require the one-step implication

$$\mathcal{P}_k(J) > F_k \implies \mathcal{P}_{k-1}(J'_{xy}) > F_{k-1} \quad \text{for some } xy \in E(J). \quad (3.4)$$

The numerical hypotheses under which we prove this implication are stated precisely in [Theorem 3.1](#).

Finally, let G be a starting graph with N vertices in each class and density exactly $1/2$; the reduction from density at least $1/2$ is justified in [Section 7.3](#). We require

$$\mathcal{P}_t(G) \geq a_0 N 2^{-t} \quad \text{and} \quad F_k \leq F_* \quad \text{for every } k,$$

where $a_0 > 0$ and F_* are absolute constants. If $N > (F_*/a_0)2^t$, then

$$\mathcal{P}_t(G) \geq a_0 N 2^{-t} > F_* \geq F_t.$$

At each stage, either the preliminary criterion finishes the construction or (3.4) reduces the number of vertices still needed by two. If the process reaches $k = 1$, the positive potential guarantees an edge. Adding back the previously chosen endpoints gives a $K_{t,t}$. This proves (3.3) by induction once the one-step implication has been established in the case where the preliminary tests fail.

The uniform bound $F_k \leq F_*$ makes the sufficient starting size a constant multiple of 2^t . If instead the thresholds grew proportionally to k , the same argument would require a starting size of order $t2^t$.

We choose nonnegative losses ℓ_k with $\sum_{k \geq 2} \ell_k < \infty$, and uniformly bounded thresholds F_k satisfying $F_k - F_{k-1} \geq \ell_k$. In the case requiring the one-step estimate, it suffices to prove the stronger expectation bound

$$\mathbb{E}_{\text{edge}} \mathcal{P}_{k-1}(J'_{xy}) \geq \mathcal{P}_k(J) - \ell_k > F_k - \ell_k \geq F_{k-1}. \quad (3.5)$$

Since the average residual potential exceeds F_{k-1} , some edge satisfies (3.4). Here

$$\mathbb{E}_{\text{edge}} W_{xy} = \frac{1}{e(J)} \sum_{xy \in E(J)} W_{xy}$$

is expectation over a uniformly chosen edge of J . A fixed loss of 2 per step would instead give $F_k = F_1 + 2(k-1)$ and lead only to a bound of order $t2^t$.

3.3 The form of the potential function

All logarithms here and throughout the paper are natural unless otherwise specified. Our goal in this section is to explain how we came to our potential function – which is chosen to be a product of a size factor and a density factor – and then to derive its dependence on k .

Let the current sizes of the vertex parts of the allowable vertices be m, n , let the current density be ρ , and suppose that k vertices remain to be chosen on each side. A potential need not have the form $f(m, n)g_k(\rho)$: it could depend jointly on m, n, k, ρ , or on further features of the graph. However, the product form appears to be a natural simple choice that works for our proof.

The function $f(m, n)$ measures the effective supply of vertices, while $g_k(\rho)$ accounts for the density of the graph. Separating these roles lets us use one size function at every stage and change the density factor as k decreases. It also has the following useful property: for positive factors,

$$\log(f(m, n)g_k(\rho)) = \log f(m, n) + \log g_k(\rho).$$

We can therefore combine a logarithmic size estimate with a logarithmic density estimate, even though the size and density of a link depend on the same chosen edge and are not independent.

We want f to measure in some form the number of available vertices, so we impose

$$f(\lambda m, \lambda n) = \lambda f(m, n), \quad f(N, N) = N.$$

The first condition says that scaling both classes scales their effective size by the same factor. Symmetry in m, n and comparability with $\min\{m, n\}$ are also natural requirements, since the smaller class limits a balanced biclique. Ignore the deletion of the selected vertices for the ensuing discussion. In an ideal step of density ρ , both class sizes are multiplied by ρ , the density remains ρ , and k decreases to $k - 1$. Before the step, the potential is $f(m, n)g_k(\rho)$. The step multiplies the size factor by ρ , but also reduces the target from $K_{k,k}$ to $K_{k-1,k-1}$. To balance these changes and preserve a large potential for the next stage, we choose $g_k(\rho) = \rho g_{k-1}(\rho)$. The potential is then unchanged in this ideal model. Iterating the recurrence gives

$$g_k(\rho) = \rho^k g_0(\rho).$$

This motivates the factor ρ^k ; the actual proof will allow small, controlled losses at each step. We therefore obtain

$$\mathcal{P}_k(J) = f(m, n)\rho^k w(\rho), \tag{3.6}$$

where w is independent of k . The exponent k records the k neighborhood restrictions still to come. At starting density $1/2$ and with equal classes, the potential is

$$\mathcal{P}_t(G) = N2^{-t}w(1/2).$$

The correction w remains to be chosen so that the actual losses, including deletion and changes in density, are summable.

3.4 Deriving the harmonic mean

The aim in this section is to explain the choice of f that allows us to compute various averages over edges. The symmetry, scaling, and smaller-class requirements above leave many choices, though there is a particularly convenient one.

Write $e = e(J) = \rho mn > 0$. For an edge xy , put $u = d_J(y)$ and $v = d_J(x)$, the sizes of the full neighborhoods before deleting x, y . If m_+, n_+ count the nonisolated vertices in the two original classes, then

$$\mathbb{E}_{\text{edge}} \left(\frac{1}{u} + \frac{1}{v} \right) = \frac{m_+ + n_+}{e(J)}. \tag{3.7}$$

Indeed, a vertex of positive degree d is an endpoint of exactly d edges, and contributes $1/d$ on each one. The degree bias in sampling an endpoint of a uniform edge therefore cancels when its degree is inverted. We will need a lower bound on the average of $\log f(u, v)$. Since concavity of the logarithm yields

$$\frac{1}{e} \sum_{xy \in E(J)} \log f(u, v) \geq -\log \left(\frac{1}{e} \sum_{xy \in E(J)} \frac{1}{f(u, v)} \right),$$

it suffices to bound the average of $1/f(u, v)$ from above. The identity (3.7) makes this particularly simple if we choose $1/f(m, n)$ to be linear in $1/m$ and $1/n$. Within this reciprocal-linear family, symmetry requires equal coefficients, and $f(N, N) = N$ fixes each coefficient to be $1/2$. We have thus arrived at the harmonic mean:

$$f(m, n) = H(m, n) = \frac{2mn}{m+n} \quad (m, n > 0). \quad (3.8)$$

If either class is empty, set $f(m, n) = H(m, n) = 0$. It satisfies all the desired size conditions:

$$\min\{m, n\} \leq H(m, n) \leq 2 \min\{m, n\}, \quad H(N, N) = N, \quad H(\lambda m, \lambda n) = \lambda H(m, n).$$

In particular, unequal candidate classes can be retained without discarding vertices merely to balance them. Write $e = e(J) = \rho mn$. For an edge xy , with $x \in L$ and $y \in R$, the full link has vertex classes $N_J(y)$ and $N_J(x)$, of sizes $d_J(y)$ and $d_J(x)$, respectively. Since $1/H(s, t) = \frac{1}{2}(1/s + 1/t)$, we have

$$\frac{1}{e} \sum_{xy \in E(J)} \frac{1}{H(d_J(y), d_J(x))} = \frac{1}{2e} \sum_{xy \in E(J)} \left(\frac{1}{d_J(x)} + \frac{1}{d_J(y)} \right) = \frac{m_+ + n_+}{2e} \leq \frac{m+n}{2\rho mn} = \frac{1}{\rho H(m, n)}.$$

Here m_+, n_+ count the nonisolated vertices in the two classes. Each such vertex of degree d contributes $1/d$ on each of its d incident edges, giving a total contribution of one. Applying concavity of the logarithm to the positive numbers $1/H(d_J(y), d_J(x))$ therefore gives

$$\frac{1}{e} \sum_{xy \in E(J)} \log H(d_J(y), d_J(x)) \geq -\log \left(\frac{1}{e} \sum_{xy \in E(J)} \frac{1}{H(d_J(y), d_J(x))} \right) \geq \log(\rho H(m, n)). \quad (3.9)$$

These calculations show why the harmonic mean is perhaps the most natural candidate for $f(n, m)$.

3.5 Logarithms and linearity of expectation

Our goal is to combine the size and density estimates for the chosen edge and retain the gain needed to pay for later errors. Given an edge xy write $q = q_{xy}$ for the density of its full link. The current density ρ and the link density q need not agree. For example, C_8 with four vertices in each class has density $1/2$; every full link has density $3/4$, whereas deletion of the chosen endpoints leaves a graph of density zero.

The size $H(u, v)$ and density q both depend on the chosen edge. Separate lower bounds on their arithmetic means cannot simply be multiplied to bound the mean of their product. Logarithms resolve this difficulty via

$$\log(H(u, v)q^r) = \log H(u, v) + r \log q,$$

as expectations of the two terms can be added without any independence assumption. One important question is therefore whether $\mathbb{E}_{\text{edge}} \log q$ is at least $\log \rho$.

Define the *logarithmic gain*

$$\delta = \mathbb{E}_{\text{edge}} \log(q/\rho) = \log \left(\frac{\exp(\mathbb{E}_{\text{edge}} \log q)}{\rho} \right). \quad (3.10)$$

The numerator in the rightmost expression is the geometric mean of the full-link densities. Thus e^δ is the factor by which this geometric mean exceeds the original density. Using (3.10), linearity of expectation, and (3.9), we obtain, for $r \geq 0$,

$$\begin{aligned} \mathbb{E}_{\text{edge}} \log(H(u, v)q^r) &= \mathbb{E}_{\text{edge}} \log H(u, v) + r \mathbb{E}_{\text{edge}} \log q \\ &\geq \log(\rho H(m, n)) + r(\log \rho + \delta) = \log(H(m, n)\rho^{r+1}) + r\delta. \end{aligned} \quad (3.11)$$

We will prove in [Section 4](#) that $\delta \geq 0$, even though individual links can have density below ρ . Thus $r\delta$ is a nonnegative gain for every $r \geq 0$. Nonnegativity alone suffices for the preliminary induction, but the refined argument that we will need in the proof needs quantitative control of the errors. Hence, the proof will also establish (4.6), which bounds explicit nonnegative sums measuring degree and common-neighbor irregularity by δ . In [Section 6](#), we use these bounds to prove (3.16): both the signed mean of $q - \rho$ and its second moment are at most $10^4 \rho^{-5} \delta$.

These moment bounds make the averaged density error at most a calculable coefficient times δ . The later argument checks that this coefficient is at most r , allowing the gain $r\delta$ to absorb the error.

3.6 The warm-up: recovering the usual bound

We first show that the uncorrected potential (i.e., without the term $w(\rho)$) recovers the classical bound, and identify the loss responsible for its factor of t .

First take $w \equiv 1$, and write

$$B_k(J) = H(m, n) \rho^k.$$

If either class is empty, set $B_k(J) = 0$ without assigning a density.

The full link contains the selected vertices x, y . Before continuing the algorithm, we must delete both vertices, together with all edges incident to them, so that neither vertex can be selected again. The resulting graph J'_{xy} has class sizes $u - 1, v - 1$. Let q' denote its density when both classes are nonempty.

We show that this deletion reduces the uncorrected potential by at most 2. Suppose first that $u, v \geq 2$ and $k \geq 2$. Within the full link, x is adjacent to all v vertices in the opposite class, and y is adjacent to all u vertices in its opposite part. Since the edge xy belongs to both stars, deleting the two vertices removes exactly $u + v - 1$ edges. Therefore

$$uvq = (u + v - 1) + (u - 1)(v - 1)q',$$

or equivalently,

$$q = \frac{u + v - 1}{uv} \cdot 1 + \frac{(u - 1)(v - 1)}{uv} q'.$$

The two coefficients are nonnegative and sum to one. Thus q is a convex combination of 1 and q' . Convexity of $z \mapsto z^{k-1}$ gives

$$q^{k-1} \leq \frac{u + v - 1}{uv} + \frac{(u - 1)(v - 1)}{uv} (q')^{k-1}.$$

Multiplying by $H(u, v) = 2uv/(u + v)$, we obtain

$$\begin{aligned} H(u, v) q^{k-1} &\leq \frac{2(u + v - 1)}{u + v} + \frac{2(u - 1)(v - 1)}{u + v} (q')^{k-1} \\ &\leq 2 + \frac{2(u - 1)(v - 1)}{u + v - 2} (q')^{k-1} = H(u - 1, v - 1) (q')^{k-1} + 2 = B_{k-1}(J'_{xy}) + 2. \end{aligned}$$

If $u = 1$ or $v = 1$, the full link is a star, so $q = 1$ and $H(u, v) < 2$. The residual graph has an empty class and hence potential zero. The same inequality therefore holds in this case as well. The deletion loss is consequently at most 2, independently of the class sizes, the density, and the number of remaining steps.

Taking $r = k - 1$ in (3.11), and using the arithmetic–geometric mean inequality with $\delta \geq 0$, gives $\mathbb{E}_{\text{edge}} H(u, v) q^{k-1} \geq \exp(\log B_k(J)) = B_k(J)$. Hence

$$\mathbb{E}_{\text{edge}} B_{k-1}(J'_{xy}) \geq B_k(J) - 2. \quad (3.12)$$

Starting with $C_1 = 0$ and $C_k = C_{k-1} + 2$, induction gives

$$B_k(J) > C_k = 2(k-1) \implies J \text{ contains } K_{k,k}.$$

Indeed, $B_1(J) > 0$ guarantees an edge. For $k \geq 2$, (3.12) gives an edge whose residual graph has $B_{k-1} > 2(k-2)$, and the induction hypothesis supplies a $K_{k-1,k-1}$ that extends to $K_{k,k}$. At density $1/2$ with equal classes, this criterion guarantees a $K_{k,k}$ when $N2^{-k} > 2(k-1)$. The factor of order k comes from accumulating a constant loss of 2 over $k-1$ steps.

3.7 Making the loss summable

We next choose the density correction so that the losses add to a constant instead of a growing function with the number of steps. The sharper bound that we will prove, for $r \geq 1$, is

$$H(u, v) q^r \leq H(u-1, v-1) (q')^r + \min\{2, 5r q^{r-1}\}. \quad (3.13)$$

For $q = 1/2$ and $r = k-1$, the second error term is $5(k-1)2^{-(k-2)}$, whose sum over $k \geq 2$ is 20. More generally, the error is summable when q stays below a fixed constant less than one.

The obstruction to summing the deletion losses is that the density q may approach one, so the bound $5r q^{r-1}$ need not be small. High density helps in finding a biclique, but the number of available vertices on each side also matters. We therefore introduce the weight

$$w(\rho) = [\rho(1-\rho)]^M$$

for a fixed positive integer M , and use the corrected potential $B_k(J)w(\rho)$. This weight has the fixed positive value $w(1/2) = 4^{-M}$ and tends to zero as ρ approaches either endpoint.

The preliminary criterion already gives a $K_{k,k}$ if $B_k(J) > 2(k-1)$. We may therefore concentrate on graphs satisfying the upper bound $B_k(J) \leq 2(k-1)$. We seek a one-step estimate when the corrected potential $B_k(J)w(\rho)$ exceeds F_k , where $F_k \geq A$ and $A > 0$ is independent of k (we will take $A = 2^{40}$). In this case,

$$A < B_k(J)w(\rho) \leq 2(k-1)w(\rho) \leq 2k w(\rho),$$

and hence

$$[\rho(1-\rho)]^M > \frac{A}{2k}. \quad (3.14)$$

Although $w(\rho)$ is small, the upper bound $2(k-1)w(\rho)$ grows with k , so these conditions are not incompatible for all k . Conversely, if (3.14) fails while $B_k(J) \leq 2(k-1)$, then

$$B_k(J)w(\rho) \leq 2(k-1) \frac{A}{2k} = A \left(1 - \frac{1}{k}\right) < A \leq F_k. \quad (3.15)$$

Thus a graph with potential above its threshold either satisfies the preliminary criterion or lies in the range (3.14). For fixed k , this range excludes densities sufficiently close to 0 or 1. The refined one-step estimate in this range will allow the thresholds F_k to remain bounded as k grows. Its complete hypotheses also include numerical bounds on the residual graphs, as stated in Theorem 3.1.

Controlling density changes is the main technical issue. In addition to $\delta \geq 0$, we will show

$$0 \leq \mathbb{E}_{\text{edge}}(q - \rho) \leq 10^4 \rho^{-5} \delta, \quad \mathbb{E}_{\text{edge}}(q - \rho)^2 \leq 10^4 \rho^{-5} \delta. \quad (3.16)$$

The upper bound on the average density change is especially important. Estimating the average density change from its second moment by Cauchy–Schwarz would give only a bound proportional to $\sqrt{\delta}$. For fixed k , this bound exceeds the available logarithmic gain of order $k\delta$ when δ is sufficiently small. We therefore estimate the average directly, retaining the signs of the individual density changes and exploiting cancellation to obtain a bound proportional to δ .

The purpose of (3.16) is to control how changes in density affect the correction factor in the potential. A Taylor estimate produces a linear term involving $\mathbb{E}_{\text{edge}}(q - \rho)$ and a quadratic error involving $\mathbb{E}_{\text{edge}}(q - \rho)^2$. The bounds above therefore make the resulting averaged error proportional to δ , with a coefficient depending on the density and the remaining number of steps. We can then compare this coefficient with that of the available logarithmic gain $r\delta$.

We prove these bounds directly in Section 6, using sums of squares that measure deviations of degrees and common neighborhoods from their expected values. These are connected with box norms and quasirandomness (see Remark 6.4). The argument requires explicit estimates linear in δ , which we establish directly in Section 6.

After a Taylor estimate and a separate bound on very dense links, the coefficient of the density error is bounded, up to a constant depending on M , by

$$\rho^{-5}(1 - \rho)^{-2} \log(2k).$$

In the range (3.14),

$$\rho^{-5}(1 - \rho)^{-2} \leq [\rho(1 - \rho)]^{-5} < \left(\frac{2k}{A}\right)^{5/M}.$$

For $M > 5$, this coefficient grows more slowly than k . Thus a sufficiently large fixed correction exponent and constant A ensure that the logarithmic gain covers the density error. The explicit proof uses $M = 20$ and $A = 2^{40}$; neither numerical choice is intended to be optimal.

3.8 The one-step density theorem

We now put all of the previously described ingredients together and state the theorem whose proof occupies the following sections. For a bipartite graph J with class sizes $m, n > 0$ and density ρ , define

$$\mathcal{P}_k(J) = H(m, n) \rho^{k+20} (1 - \rho)^{20} = B_k(J) [\rho(1 - \rho)]^{20}. \quad (3.17)$$

Set $\mathcal{P}_k(J) = 0$ if either class is empty. In particular, edgeless graphs and graphs of density one have potential zero. The three factors have separate roles: H measures the available vertices, ρ^k accounts for the remaining neighborhood steps, and $[\rho(1 - \rho)]^{20}$ controls the endpoint densities. For the thresholds, set

$$A = 2^{40}, \quad F_k = A \exp\left(\sum_{j=2}^k \frac{1}{j^2}\right). \quad (3.18)$$

Thus $F_1 = A$, $F_k = F_{k-1} e^{1/k^2}$ for $k \geq 2$, and

$$F_k < 2A = 2^{41}. \quad (3.19)$$

For the last inequality, one may use the elementary estimate

$$\sum_{j=2}^{\infty} \frac{1}{j^2} \leq \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \int_4^{\infty} x^{-2} dx = \frac{97}{144} < \log 2.$$

Theorem 3.1 (One-step density theorem). *Let $k \geq 2$ be an integer and let G be a bipartite graph. For each edge $xy \in E(G)$, let*

$$G'_{xy} = G[N_G(y) \setminus \{x\}, N_G(x) \setminus \{y\}]$$

be its residual link. Suppose that

$$B_k(G) \leq 2(k-1) \tag{3.20}$$

and

$$B_{k-1}(G'_{xy}) \leq 2(k-2) \quad \text{for every } xy \in E(G). \tag{3.21}$$

If $\mathcal{P}_k(G) > F_k$, then

$$\mathbb{E}_{\text{edge}} \mathcal{P}_{k-1}(G'_{xy}) \geq \mathcal{P}_k(G) - \frac{F_{k-1}}{k^2} > F_{k-1}. \tag{3.22}$$

In particular, some edge satisfies $\mathcal{P}_{k-1}(G'_{xy}) > F_{k-1}$.

The hypotheses involve only the sizes and densities of G and its residual links; no forbidden-subgraph assumption is imposed. If (3.20) fails, Lemma 5.2 gives a $K_{k,k}$ in G . If (3.21) fails, the same criterion gives a $K_{k-1,k-1}$ in a residual link, which extends to a $K_{k,k}$ in G .

Sections 4–6 prove the logarithmic, deletion, and density-change estimates needed for Theorem 3.1. We combine them to prove the theorem in Section 7.1, then give the complete induction and the deduction of the main results in Sections 7.2 and 7.3.

4 The full-link inequality: Proof of (3.11)

We begin the proof of Theorem 3.1 by establishing (3.11), together with the quantitative bound (4.6). The latter controls degree and common-neighbor errors and will be used in Section 6 to bound changes in the link densities.

The logarithmic averages used here are related to the logarithmic calculus of Li and Szegedy [9], developed in the study of Sidorenko's conjecture on lower bounds for homomorphism densities of bipartite graphs [10]. We give a self-contained proof, including the quantitative bound (4.6) needed below.

Let J be a bipartite graph with vertex classes L, R , where

$$|L| = m, \quad |R| = n, \quad e = |E(J)| > 0, \quad \rho = \frac{e}{mn}.$$

Write $N(x) = N_J(x)$ and $d(x) = |N(x)|$. For $X \subseteq L$ and $Y \subseteq R$, let $e_J(X, Y)$ denote the number of edges between X and Y . Graphs with no edges, including graphs with an empty class, have zero potential and do not satisfy the hypothesis $\mathcal{P}_k(G) > F_k$ of Theorem 3.1. All logarithms are natural.

For an edge ij , with $i \in L$ and $j \in R$, its *full link* is the bipartite graph with classes $N(j)$ and $N(i)$, containing every edge of J between these classes. Its density is

$$q_{ij} = \frac{e_J(N(j), N(i))}{d(j)d(i)}. \tag{4.1}$$

Since ij is an edge, $i \in N(j)$ and $j \in N(i)$. Thus the full link still contains the selected vertices and, in particular, $q_{ij} > 0$. Recall that

$$H(s, t) = \frac{2st}{s+t} \quad (s, t > 0).$$

The two part sizes of the full link of ij are $d(j)$ and $d(i)$, so their harmonic mean is $H(d(j), d(i))$.

4.1 The logarithmic gain and the error sums

Define

$$\delta = \frac{1}{e} \sum_{ij \in E(J)} \log \frac{q_{ij}}{\rho}. \quad (4.2)$$

The geometric mean of the full-link densities is therefore $\rho \exp(\delta)$. We will prove that $\delta \geq 0$ and, more precisely, that δ bounds the sum of three nonnegative quantities measuring irregularity. For $i, i' \in L$, define

$$C_{ii'} = |N(i) \cap N(i')|, \quad W_{ii'} = \sum_{j \in N(i) \cap N(i')} \frac{1}{d(j)}. \quad (4.3)$$

Thus $C_{ii'}$ counts common neighbors, while $W_{ii'}$ gives a common neighbor of smaller degree a larger weight. Every sum over $i, i' \in L$ includes all ordered pairs, including those with $i = i'$. The row sums of W have a simple graph interpretation:

$$\sum_{i' \in L} W_{ii'} = \sum_{j \in N(i)} \frac{|N(j)|}{d(j)} = d(i). \quad (4.4)$$

Consequently,

$$\sum_{i, i' \in L} W_{ii'} = e, \quad \sum_{i, i' \in L} \frac{d(i)d(i')}{e} = e.$$

These identities will allow several terms to cancel in the proof. We use the function

$$\psi(s, t) = s \log(s/t) - s + t \quad (s \geq 0, t > 0),$$

with $\psi(0, t) = t$ and $\psi(0, 0) = 0$. Its nonnegativity and quadratic lower bound are recorded in [Lemma A.2](#). Define

$$\begin{aligned} S_L &= \frac{1}{e} \sum_{i \in L} \psi\left(d(i), \frac{e}{m}\right), & S_R &= \frac{1}{e} \sum_{j \in R} \psi\left(d(j), \frac{e}{n}\right), \\ S_C &= \frac{1}{e} \sum_{i, i' \in L} \psi\left(W_{ii'}, \frac{d(i)d(i')}{e}\right). \end{aligned} \quad (4.5)$$

If $d(i)d(i') = 0$, then $W_{ii'} = 0$, so the corresponding term in S_C is well defined and equals zero. Each of the three sums is nonnegative.

The quantities S_L, S_R compare the degrees with their respective average degrees $e/m, e/n$. The quantity S_C compares the weighted common-neighbor sums with $d(i)d(i')/e$. Controlling these three quantities by δ is the quantitative information needed later.

Lemma 4.1. *With the notation above,*

$$\delta \geq S_L + S_R + S_C \geq 0. \quad (4.6)$$

Moreover, for every real $r \geq 0$,

$$\frac{1}{e} \sum_{ij \in E(J)} \log(H(d(j), d(i))q_{ij}^r) \geq \log(H(m, n)\rho^{r+1}) + r\delta. \quad (4.7)$$

Proof. In the logarithmic sums below, terms with coefficient zero are interpreted as zero. Put

$$h_L = \frac{1}{e} \sum_{i \in L} d(i) \log d(i), \quad h_R = \frac{1}{e} \sum_{j \in R} d(j) \log d(j).$$

Since the degrees in each class sum to e , expanding the definitions of S_L, S_R gives

$$S_L = h_L - \log(e/m), \quad S_R = h_R - \log(e/n). \quad (4.8)$$

For example, the linear terms in S_L cancel because

$$\sum_{i \in L} \left(-d(i) + \frac{e}{m} \right) = 0.$$

Similarly, the linear terms in S_C cancel, giving

$$S_C = \frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log \frac{e W_{ii'}}{d(i) d(i')} = \frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log W_{ii'} + \log e - 2h_L. \quad (4.9)$$

For the last equality, use $\sum_{i'} W_{ii'} = d(i)$ and the symmetry of W . In particular,

$$\frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log d(i) = \frac{1}{e} \sum_{i \in L} d(i) \log d(i) = h_L,$$

and the term containing $\log d(i')$ has the same value. We next compare $C_{ii'}$ and $W_{ii'}$. Fix i, i' with $W_{ii'} > 0$. The numbers

$$\frac{1}{d(j) W_{ii'}}, \quad j \in N(i) \cap N(i'),$$

are positive and sum to one. Applying the weighted AM–GM inequality to the degrees of these common neighbors gives

$$\log \frac{C_{ii'}}{W_{ii'}} \geq \frac{1}{W_{ii'}} \sum_{j \in N(i) \cap N(i')} \frac{\log d(j)}{d(j)}.$$

Multiplying by $W_{ii'}$, we obtain

$$W_{ii'} \log C_{ii'} \geq W_{ii'} \log W_{ii'} + \sum_{j \in N(i) \cap N(i')} \frac{\log d(j)}{d(j)}.$$

Now sum over i, i' and divide by e . Each vertex $j \in R$ is a common neighbor of exactly $d(j)^2$ ordered pairs in L . Therefore

$$\frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log C_{ii'} \geq \frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log W_{ii'} + h_R. \quad (4.10)$$

For each edge ij , count the edges in its full link by their endpoint $i' \in N(j)$:

$$e_J(N(j), N(i)) = \sum_{i' \in N(j)} C_{ii'}.$$

Hence

$$q_{ij} = \frac{1}{d(i)} \left(\frac{1}{d(j)} \sum_{i' \in N(j)} C_{ii'} \right).$$

Every term $C_{ii'}$ in this sum is positive, since j is a common neighbor of i and i' . The AM–GM inequality gives

$$\log q_{ij} \geq \frac{1}{d(j)} \sum_{i' \in N(j)} \log C_{ii'} - \log d(i).$$

Sum over all edges and divide by e . For fixed i, i' , the coefficient of $\log C_{ii'}$ is

$$\frac{1}{e} \sum_{j \in N(i) \cap N(i')} \frac{1}{d(j)} = \frac{W_{ii'}}{e}.$$

Also,

$$\frac{1}{e} \sum_{ij \in E(J)} \log d(i) = h_L.$$

It follows from (4.10), (4.9), and (4.8) that

$$\begin{aligned} \frac{1}{e} \sum_{ij \in E(J)} \log q_{ij} &\geq \frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log C_{ii'} - h_L \\ &\geq \frac{1}{e} \sum_{i, i' \in L} W_{ii'} \log W_{ii'} + h_R - h_L = S_C - \log e + h_L + h_R = \log \rho + S_L + S_R + S_C. \end{aligned}$$

Subtracting $\log \rho$ proves (4.6).

It remains to account for the class sizes of the full links. Let m_+ and n_+ be the numbers of nonisolated vertices in L and R . Since

$$\frac{1}{H(d(j), d(i))} = \frac{1}{2} \left(\frac{1}{d(j)} + \frac{1}{d(i)} \right),$$

we have

$$\frac{1}{e} \sum_{ij \in E(J)} \frac{1}{H(d(j), d(i))} = \frac{m_+ + n_+}{2e} \leq \frac{m + n}{2e} = \frac{1}{\rho H(m, n)}.$$

Indeed, each nonisolated vertex x occurs in $d(x)$ edges and contributes $1/d(x)$ for each of them. Apply concavity of the logarithm to these reciprocals:

$$\frac{1}{e} \sum_{ij \in E(J)} \log H(d(j), d(i)) \geq -\log \left(\frac{1}{e} \sum_{ij \in E(J)} \frac{1}{H(d(j), d(i))} \right) \geq \log(\rho H(m, n)).$$

Finally, (4.2) gives

$$\frac{r}{e} \sum_{ij \in E(J)} \log q_{ij} = r(\log \rho + \delta).$$

Adding this to the preceding inequality proves (4.7). $\square$

5 Deletion and the preliminary density criterion: Proof of (3.13)

We prove the deletion estimate (3.13), restated as Lemma 5.1, for use in Theorem 3.1. We also prove the preliminary density criterion, Lemma 5.2, which handles the cases where the numerical hypotheses of the one-step theorem fail.

Fix an edge xy , and abbreviate $u = |N(y)|$, $v = |N(x)|$, $q = q_{xy}$. Let J'_{xy} be the bipartite graph between $N(y) \setminus \{x\}$ and $N(x) \setminus \{y\}$, and let q' be its density. If either of these two sets is empty, set $q' = 0$ and interpret every residual potential as zero. For $k \geq 2$, every $K_{k-1, k-1}$ in J'_{xy} extends to a $K_{k, k}$ in J by adding x, y .

Lemma 5.1. *For every real $r \geq 1$,*

$$H(u, v)q^r \leq H(u-1, v-1)(q')^r + \min\{2, 5rq^{r-1}\}. \quad (5.1)$$

Proof. In the bipartite graph between $N(y)$ and $N(x)$, the vertex x is adjacent to all v vertices on the other side, and y is adjacent to all u vertices on its other side. These two stars have exactly $w = u + v - 1$ edges in their union. If $u, v \geq 2$, then

$$q = \frac{w}{uv} + \frac{(u-1)(v-1)}{uv}q'. \quad (5.2)$$

In particular, $q' \leq q$. Convexity of $z \mapsto z^r$ gives

$$H(u, v)q^r \leq \frac{2w}{u+v} + \frac{2(u-1)(v-1)}{u+v}(q')^r \leq 2 + H(u-1, v-1)(q')^r.$$

For the refined bound, write $H_0 = H(u, v)$ and $H_1 = H(u-1, v-1)$. Direct algebra gives

$$H_0 - H_1 = \frac{2(u^2 + v^2 - u - v)}{(u+v)(u+v-2)} \leq 2.$$

Also, (5.2) implies

$$H_1(q - q') = \frac{2(1-q)(u+v-1)}{u+v-2} \leq 3(1-q),$$

where the last inequality uses $u+v \geq 4$. Since $q^r - (q')^r \leq rq^{r-1}(q - q')$, we have

$$H_0q^r - H_1(q')^r = (H_0 - H_1)q^r + H_1(q^r - (q')^r) \leq 2q^r + 3r(1-q)q^{r-1} \leq 5rq^{r-1}.$$

Combining the two bounds proves (5.1) when $u, v \geq 2$.

If $u = 1$ or $v = 1$, the full link consists of one of the two stars, so $q = 1$, $H(u, v) < 2$, and the residual potential is zero. Since $\min\{2, 5r\} = 2$, the same conclusion holds. $\square$

We now give the formal proof of the preliminary density criterion from the warm-up argument.

Lemma 5.2 (Preliminary density criterion). *For every positive integer k and every bipartite graph J ,*

$$B_k(J) > 2(k-1) \implies J \text{ contains } K_{k,k}. \quad (5.3)$$

Proof. We induct on k . If $k = 1$, then $B_1(J) > 0$ implies that J has an edge. Now let $k \geq 2$ and assume $B_k(J) > 2(k-1)$. The graph has nonempty classes and positive density. For a uniformly chosen edge, let u, v, q be the full-link sizes and density. Taking $r = k-1$ in (4.7), using $\delta \geq 0$ from Lemma 4.1, and applying concavity of the logarithm, we obtain

$$\mathbb{E}_{\text{edge}} H(u, v)q^{k-1} \geq B_k(J).$$

By Lemma 5.1,

$$\mathbb{E}_{\text{edge}} B_{k-1}(J'_{xy}) \geq B_k(J) - 2 > 2(k-2).$$

Some edge therefore has $B_{k-1}(J'_{xy}) > 2(k-2)$. The induction hypothesis gives a $K_{k-1, k-1}$ in that residual graph, and adding the selected endpoints gives a $K_{k,k}$ in J . $\square$

6 Controlling the link densities: Proof of (3.16)

The purpose of this section is to prove (3.16), restated below as Lemma 6.1. These estimates control the density errors in the proof of Theorem 3.1. Let J have vertex classes L, R of sizes m, n , with $e = |E(J)| > 0$ and $\rho = e/(mn)$. Write $d(x) = |N_J(x)|$. We retain the full-link densities q_{ij} , the logarithmic gain δ , and the quantities S_L, S_R, S_C from Section 4. In sums without specified ranges, i, i' run over L and j, j' over R . Indices from the same class need not be distinct.

Lemma 6.1. *For every bipartite graph with $\rho > 0$,*

$$0 \leq \frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho) \leq 10^4 \rho^{-5} \delta, \quad \frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho)^2 \leq 10^4 \rho^{-5} \delta. \quad (6.1)$$

6.1 Degrees, common neighbors, and four-cycles

The average degrees on L and R are e/m and e/n . Define their sums of squared deviations by

$$D_L = \sum_i \left(d(i) - \frac{e}{m} \right)^2, \quad D_R = \sum_j \left(d(j) - \frac{e}{n} \right)^2. \quad (6.2)$$

Lemma 6.2. *We have*

$$D_L \leq 2neS_L, \quad D_R \leq 2meS_R, \quad mD_L + nD_R \leq 2mne\delta. \quad (6.3)$$

Proof. Since $d(i) \leq n$ and $0 < e/m \leq n$, apply (A.3) with $s = d(i)$, $t = e/m$, and $M = n$. Summing the resulting inequalities gives

$$D_L \leq 2n \sum_i \psi \left(d(i), \frac{e}{m} \right) = 2neS_L.$$

The right-side estimate follows with $M = m$. Multiplying these bounds by m, n , respectively, and using $S_L + S_R \leq \delta$ proves the last inequality. $\square$

We next control the common-neighbor counts $C_{ii'} = |N(i) \cap N(i')|$ from (4.3). For each ordered pair i, i' , choosing an ordered pair of common neighbors j, j' gives the closed walk i, j, i', j', i . The same walk can be specified by an edge ij and an edge $i'j'$ of its full link. Counting these ordered choices in the two ways gives

$$\sum_{i, i'} C_{ii'}^2 = \sum_{ij \in E(J)} e_J(N(j), N(i)). \quad (6.4)$$

Thus four-cycle counts arise naturally when we sum the edge counts of full links. Here and below these are labeled four-cycle counts allowing repeated vertices.

Regularity alone does not force this count to be close to $\rho^4 m^2 n^2$. For example, two disjoint copies of $K_{N/2, N/2}$ form a bipartite graph with N vertices in each class, density $1/2$, and every degree equal to $N/2$. Nevertheless, the sum in (6.4) is $N^4/8$, twice the benchmark $\rho^4 N^4 = N^4/16$. We therefore need the common-neighbor information in S_C as well as the degree information in S_L, S_R .

Let A_{ij} be the adjacency indicator. Define

$$B_{ii'} = C_{ii'} - \rho d(i) - \rho d(i') + \rho^2 n = \sum_j (A_{ij} - \rho)(A_{i'j} - \rho), \quad R_4 = \sum_{i, i'} B_{ii'}^2. \quad (6.5)$$

The subtraction in $B_{ii'}$ centers the adjacency entries; expanding the square shows that

$$R_4 = \sum_{i,i',j,j'} (A_{ij} - \rho)(A_{i'j} - \rho)(A_{ij'} - \rho)(A_{i'j'} - \rho).$$

If every left degree is ρn and every right degree is ρm , then $B_{ii'} = C_{ii'} - \rho^2 n$ and $\sum_{i,i'} C_{ii'} = \sum_j d(j)^2 = \rho^2 m^2 n$. Consequently,

$$R_4 = \sum_{i,i'} C_{ii'}^2 - \rho^4 m^2 n^2.$$

In this special case R_4 is exactly the excess four-cycle count. The following bound also accounts for unequal degrees.

Lemma 6.3. *We have*

$$R_4 \leq 16e^2 S_C + 20nD_R + \frac{2D_L^2}{n^2} \leq 60m^2 n^2 \delta. \quad (6.6)$$

Proof. Recall the weighted common-neighbor sums

$$W_{ii'} = \sum_{j \in N(i) \cap N(i')} \frac{1}{d(j)}.$$

Put

$$\hat{C}_{ii'} = \frac{e}{n} W_{ii'}, \quad K_{ii'} = C_{ii'} - \hat{C}_{ii'}.$$

The factor e/n is the average right degree. Thus $\hat{C}_{ii'} = C_{ii'}$ when all right degrees are equal. In general,

$$K_{ii'} = \sum_{j \in N(i) \cap N(i')} \left(1 - \frac{e}{nd(j)}\right).$$

Let $R_+ = \{j \in R : d(j) > 0\}$. Squaring and changing the order of summation gives

$$\sum_{i,i'} K_{ii'}^2 = \sum_{j,j' \in R_+} \left(1 - \frac{e}{nd(j)}\right) \left(1 - \frac{e}{nd(j')}\right) |N(j) \cap N(j')|^2.$$

Since $|N(j) \cap N(j')|^2 \leq d(j)d(j')$, taking absolute values of the coefficients and applying Cauchy–Schwarz yields

$$\sum_{i,i'} K_{ii'}^2 \leq \left(\sum_{j \in R_+} \left| d(j) - \frac{e}{n} \right| \right)^2 \leq n \sum_{j \in R_+} \left(d(j) - \frac{e}{n} \right)^2 \leq nD_R. \quad (6.7)$$

By the identity $\psi(cs, ct) = c\psi(s, t)$ for $c > 0$ and the definition of S_C ,

$$\sum_{i,i'} \psi \left(\hat{C}_{ii'}, \frac{d(i)d(i')}{n} \right) = \frac{e}{n} \sum_{i,i'} \psi \left(W_{ii'}, \frac{d(i)d(i')}{e} \right) = \frac{e^2}{n} S_C.$$

For pairs with $\hat{C}_{ii'} \leq 2n$, both arguments of ψ are at most $2n$. Hence (A.3) gives

$$\left(\hat{C}_{ii'} - \frac{d(i)d(i')}{n} \right)^2 \leq 4n \psi \left(\hat{C}_{ii'}, \frac{d(i)d(i')}{n} \right).$$

If $d(i)d(i') = 0$, both sides are zero. For pairs with $\widehat{C}_{ii'} > 2n$, the bounds $C_{ii'} \leq n$ and $d(i)d(i')/n \leq n$ give

$$|K_{ii'}| \geq \frac{\widehat{C}_{ii'}}{2}, \quad \left(\widehat{C}_{ii'} - \frac{d(i)d(i')}{n} \right)^2 \leq \widehat{C}_{ii'}^2 \leq 4K_{ii'}^2.$$

Summing over these two sets of pairs and using (6.7), we obtain

$$\sum_{i,i'} \left(\widehat{C}_{ii'} - \frac{d(i)d(i')}{n} \right)^2 \leq 4e^2 S_C + 4nD_R. \quad (6.8)$$

As $C_{ii'} = K_{ii'} + \widehat{C}_{ii'}$, the inequality $(x+y)^2 \leq 2x^2 + 2y^2$ now gives

$$\sum_{i,i'} \left(C_{ii'} - \frac{d(i)d(i')}{n} \right)^2 \leq 8e^2 S_C + 10nD_R. \quad (6.9)$$

Finally,

$$B_{ii'} = C_{ii'} - \frac{d(i)d(i')}{n} + \frac{1}{n} \left(d(i) - \frac{e}{m} \right) \left(d(i') - \frac{e}{m} \right).$$

Applying the same inequality once more gives

$$R_4 \leq 16e^2 S_C + 20nD_R + \frac{2D_L^2}{n^2}.$$

Since $e \leq mn$, $D_L \leq mn^2$, and $S_L, S_R, S_C \leq \delta$, Lemma 6.2 bounds the three terms by $16m^2n^2\delta$, $40m^2n^2\delta$, and $4m^2n^2\delta$, respectively. This proves (6.6). $\square$

Remark 6.4. The ratio $R_4/(m^2n^2)$ is the fourth power of the rectangular box norm of $A - \rho$, the centered four-cycle expression related to quasirandomness [2, 5]. Here (6.6) supplies the linear dependence on δ needed below.

6.2 Controlling full-link edge counts

The full link of an edge ij has $d(i)d(j)$ possible edges. To compare its density with ρ , we therefore compare its edge count with $\rho d(i)d(j)$. In this section, we use the degree and common-neighbor estimates from the previous section to bound both the sum of squared edge-count errors and their signed sum over edges. Both bounds are linear in δ .

These estimates are the next step towards controlling $q_{ij} - \rho$: the following subsection divides the edge-count errors by $d(i)d(j)$ and controls the effect of these varying denominators.

Lemma 6.5. *For every $i \in L$ and $j \in R$, define*

$$T_{ij} = e_J(N(j), N(i)) - \rho d(i)d(j). \quad (6.10)$$

Then

$$\sum_{i,j} T_{ij}^2 \leq 186m^3n^3\delta, \quad \sum_{ij \in E(J)} T_{ij} \leq 65m^2n^2\delta. \quad (6.11)$$

On every edge ij ,

$$q_{ij} - \rho = \frac{T_{ij}}{d(i)d(j)}, \quad (6.12)$$

and for every pair i, j ,

$$|T_{ij}| \leq d(i)d(j) \leq mn. \quad (6.13)$$

Proof. The definition of the full-link density gives (6.12). Also, $0 \leq e_J(N(j), N(i)) \leq d(i)d(j)$, so both terms in (6.10) lie between zero and $d(i)d(j)$. This proves (6.13).

To prove the two sum bounds, put

$$\xi_i = \sum_{j \in N(i)} \left(d(j) - \frac{e}{n} \right), \quad \eta_j = \sum_{i \in N(j)} \left(d(i) - \frac{e}{m} \right), \quad D_{ij} = \sum_{i'} B_{ii'} (A_{i'j} - \rho).$$

The first two quantities sum degree deviations over a neighborhood. Counting common neighbors first gives

$$\sum_{i'} C_{ii'} = \sum_{j \in N(i)} d(j), \quad \sum_{i'} B_{ii'} = \xi_i.$$

Moreover, $e_J(N(j), N(i)) = \sum_{i' \in N(j)} C_{ii'}$. Substituting the definition of $B_{ii'}$ gives

$$T_{ij} = \sum_{i' \in N(j)} B_{ii'} + \rho \eta_j = \rho \xi_i + \rho \eta_j + D_{ij}. \quad (6.14)$$

By Cauchy–Schwarz on each neighborhood,

$$\sum_i \xi_i^2 \leq \sum_i d(i) \sum_{j \in N(i)} \left(d(j) - \frac{e}{n} \right)^2 \leq n \sum_j d(j) \left(d(j) - \frac{e}{n} \right)^2 \leq mn D_R.$$

Similarly, $\sum_j \eta_j^2 \leq mn D_L$. For D_{ij} , Cauchy–Schwarz gives

$$D_{ij}^2 \leq \left(\sum_{i'} B_{ii'}^2 \right) \left(\sum_{i'} (A_{i'j} - \rho)^2 \right).$$

The factors depend only on i and j , respectively. Since

$$\sum_{i,j} (A_{ij} - \rho)^2 = e(1 - \rho) \leq mn, \quad (6.15)$$

summing the preceding inequality yields

$$\sum_{i,j} D_{ij}^2 \leq \left(\sum_{i,i'} B_{ii'}^2 \right) \left(\sum_{i',j} (A_{i'j} - \rho)^2 \right) \leq mn R_4.$$

Consequently, (6.14) and $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ imply

$$\begin{aligned} \sum_{i,j} T_{ij}^2 &\leq 3\rho^2 \left(n \sum_i \xi_i^2 + m \sum_j \eta_j^2 \right) + 3 \sum_{i,j} D_{ij}^2 \leq 3mn(\rho^2(mD_L + nD_R) + R_4) \\ &\leq 6\rho^2 m^2 n^2 e \delta + 180m^3 n^3 \delta \leq 186m^3 n^3 \delta. \end{aligned}$$

Here we used Lemmas 6.2 and 6.3 and $e \leq mn$. For the signed sum, define the mixed degree error

$$\Gamma = \sum_{ij \in E(J)} \left(d(i) - \frac{e}{m} \right) \left(d(j) - \frac{e}{n} \right) = \sum_i \left(d(i) - \frac{e}{m} \right) \xi_i = \sum_j \left(d(j) - \frac{e}{n} \right) \eta_j. \quad (6.16)$$

Changing the order of summation gives

$$\sum_i \xi_i = \sum_j d(j) \left(d(j) - \frac{e}{n} \right) = D_R, \quad \sum_j \eta_j = D_L.$$

It follows that

$$\sum_{ij \in E(J)} \xi_i = \frac{e}{m} D_R + \Gamma, \quad \sum_{ij \in E(J)} \eta_j = \frac{e}{n} D_L + \Gamma.$$

Also, by the definitions of D_{ij} and $B_{ii'}$,

$$\sum_{ij \in E(J)} D_{ij} = \sum_{i, i'} B_{ii'} (C_{ii'} - \rho d(i)) = R_4 + \rho \sum_{i'} \left(d(i') - \frac{e}{m} \right) \sum_i B_{ii'} = R_4 + \rho \Gamma.$$

For the last equality, use the symmetry of B and $\sum_i B_{ii'} = \xi_{i'}$. Summing (6.14) over the edges therefore gives the exact identity

$$\sum_{ij \in E(J)} T_{ij} = \rho^2 (m D_L + n D_R) + 3\rho \Gamma + R_4. \quad (6.17)$$

Finally, Cauchy–Schwarz and the bound on $\sum_i \xi_i^2$ give

$$|\Gamma| \leq \sqrt{D_L \sum_i \xi_i^2} \leq \sqrt{mn D_L D_R} \leq \frac{m D_L + n D_R}{2} \leq mne\delta.$$

Substitution into (6.17) yields

$$\sum_{ij \in E(J)} T_{ij} \leq 2\rho^2 mne\delta + 3\rho mne\delta + 60m^2 n^2 \delta \leq 65m^2 n^2 \delta,$$

as required. $\square$

6.3 Completing the proof of Lemma 6.1

The preceding subsection controls the edge-count errors T_{ij} . To obtain the required density estimates, we use (6.12):

$$q_{ij} - \rho = \frac{T_{ij}}{d(i)d(j)}.$$

The remaining difficulty is that the denominator varies with the edge and may be small. We first use the degree estimates to control the contribution of edges with a small endpoint degree. On the remaining edges, a lower bound on $d(i)d(j)$ suffices to bound the sum of squared density errors.

The signed sum requires more care. We compare $1/(d(i)d(j))$ with the constant mn/e^2 , the reciprocal of the product of the average degrees. This allows us to use the signed-sum bound from Lemma 6.5 and then control the error introduced by replacing the denominator.

Proof of Lemma 6.1. Recall that we must show

$$0 \leq \frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho) \leq 10^4 \rho^{-5} \delta, \quad \frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho)^2 \leq 10^4 \rho^{-5} \delta.$$

We first bound the sum of squares, then prove the upper and lower bounds for the signed sum. Let $\mathcal{B}$ be the set of edges ij with $d(i) < e/(2m)$ or $d(j) < e/(2n)$. There are at most $4m^2 D_L/e^2$ left vertices with $d(i) < e/(2m)$, since each contributes more than $e^2/(4m^2)$ to D_L . Each has fewer than $e/(2m)$ incident edges, so together they meet at most $2m D_L/e$ edges. The corresponding right-side count is at most $2n D_R/e$. Thus

$$|\mathcal{B}| \leq \frac{2(m D_L + n D_R)}{e} \leq 4mn\delta, \quad \frac{|\mathcal{B}|}{e} \leq 4\rho^{-1}\delta. \quad (6.18)$$

Outside $\mathcal{B}$, we have $d(i)d(j) \geq e^2/(4mn)$. Since $|q_{ij} - \rho| \leq 1$ on every edge, (6.12) and (6.11) give

$$\begin{aligned} \frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho)^2 &\leq \frac{|\mathcal{B}|}{e} + \frac{1}{e} \sum_{ij \in E(J) \setminus \mathcal{B}} \frac{T_{ij}^2}{d(i)^2 d(j)^2} \\ &\leq 4\rho^{-1}\delta + \frac{16m^2 n^2}{e^5} \sum_{i,j} T_{ij}^2 \leq 4\rho^{-1}\delta + 2976\rho^{-5}\delta \leq 2980\rho^{-5}\delta. \end{aligned} \quad (6.19)$$

For the signed sum, applying Cauchy–Schwarz directly to (6.19) would give a bound proportional to $\sqrt{\delta}$. We split the signed average into contributions from $\mathcal{B}$ and its complement $\mathcal{G} = E(J) \setminus \mathcal{B}$. Define

$$M_{\mathcal{B}} = \frac{1}{e} \sum_{ij \in \mathcal{B}} (q_{ij} - \rho), \quad M_{\mathcal{G}} = \frac{1}{e} \sum_{ij \in \mathcal{G}} (q_{ij} - \rho).$$

Thus

$$\frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho) = M_{\mathcal{B}} + M_{\mathcal{G}}.$$

We bound these two contributions separately. First, on $\mathcal{B}$, the bound $|q_{ij} - \rho| \leq 1$ and (6.18) give

$$|M_{\mathcal{B}}| \leq \frac{|\mathcal{B}|}{e} \leq 4\rho^{-1}\delta.$$

Next, consider $M_{\mathcal{G}}$. By (6.12),

$$M_{\mathcal{G}} = \frac{1}{e} \sum_{ij \in \mathcal{G}} \frac{T_{ij}}{d(i)d(j)}.$$

We compare $d(i)d(j)$ with the product of the average degrees, $e^2/(mn)$, and write

$$M_{\mathcal{G}} = \frac{mn}{e^3} \sum_{ij \in \mathcal{G}} T_{ij} + R_{\mathcal{G}},$$

where

$$R_{\mathcal{G}} = \frac{1}{e} \sum_{ij \in \mathcal{G}} T_{ij} \left(\frac{1}{d(i)d(j)} - \frac{mn}{e^2} \right).$$

We now bound these two terms. The signed-sum estimate in (6.11) applies to all edges. To use it for $\mathcal{G}$, observe that

$$\sum_{ij \in \mathcal{G}} T_{ij} = \sum_{ij \in E(J)} T_{ij} - \sum_{ij \in \mathcal{B}} T_{ij}.$$

Since $|T_{ij}| \leq mn$ and $|\mathcal{B}| \leq 4mn\delta$,

$$\frac{mn}{e^3} \sum_{ij \in \mathcal{G}} T_{ij} \leq \frac{mn}{e^3} (65m^2 n^2 \delta + mn|\mathcal{B}|) \leq 69\rho^{-3}\delta.$$

It remains to bound $R_{\mathcal{G}}$. For $ij \in \mathcal{G}$, we have $d(i)d(j) \geq e^2/(4mn)$, so

$$\left| \frac{1}{d(i)d(j)} - \frac{mn}{e^2} \right| \leq \frac{4m^2 n^2}{e^4} \left| d(i)d(j) - \frac{e^2}{mn} \right|.$$

Using

$$d(i)d(j) - \frac{e^2}{mn} = d(i) \left(d(j) - \frac{e}{n} \right) + \frac{e}{n} \left(d(i) - \frac{e}{m} \right),$$

together with $d(i) \leq n$ and $e/n \leq m$, we obtain

$$\left| d(i)d(j) - \frac{e^2}{mn} \right| \leq m \left| d(i) - \frac{e}{m} \right| + n \left| d(j) - \frac{e}{n} \right|.$$

By [Lemma 6.2](#) and $e \leq mn$,

$$\sum_{i,j} \left(m \left| d(i) - \frac{e}{m} \right| + n \left| d(j) - \frac{e}{n} \right| \right)^2 \leq 2mn(mD_L + nD_R) \leq 4m^3n^3\delta.$$

Here the sum is over all pairs in $L \times R$. Applying the reciprocal bound and extending the resulting nonnegative sum from $\mathcal{G}$ to all such pairs gives

$$|R_{\mathcal{G}}| \leq \frac{4m^2n^2}{e^5} \sum_{i,j} |T_{ij}| \left(m \left| d(i) - \frac{e}{m} \right| + n \left| d(j) - \frac{e}{n} \right| \right).$$

Cauchy–Schwarz and [\(6.11\)](#) now yield

$$|R_{\mathcal{G}}| \leq \frac{4m^2n^2}{e^5} \sqrt{186m^3n^3\delta \cdot 4m^3n^3\delta} \leq 112\rho^{-5}\delta.$$

Consequently,

$$M_{\mathcal{G}} \leq 69\rho^{-3}\delta + 112\rho^{-5}\delta.$$

Combining the bounds for $M_{\mathcal{B}}$ and $M_{\mathcal{G}}$ gives

$$\frac{1}{e} \sum_{ij \in E(J)} (q_{ij} - \rho) \leq 4\rho^{-1}\delta + 69\rho^{-3}\delta + 112\rho^{-5}\delta \leq 185\rho^{-5}\delta. \quad (6.20)$$

Finally, [Lemma A.1](#) and [\(4.6\)](#) give

$$\frac{1}{e} \sum_{ij \in E(J)} q_{ij} \geq \exp \left(\frac{1}{e} \sum_{ij \in E(J)} \log q_{ij} \right) = \rho \exp(\delta) \geq \rho.$$

This proves the lower bound in [\(6.1\)](#); its two upper bounds follow from [\(6.19\)](#) and [\(6.20\)](#). □

7 Proof of the one-step density theorem

We now combine the estimates from Sections 4–6 to prove [Theorem 3.1](#), with the potential and thresholds defined in [\(3.17\)](#) and [\(3.18\)](#). After proving the theorem, we deduce a density criterion for arbitrary bipartite graphs and then the main results.

7.1 Verifying the one-step inequality

Proof of Theorem 3.1. Let m, n, ρ be the class sizes and density of G , and write $e = e(G)$. Since $\mathcal{P}_k(G) > F_k$, we have $e > 0$ and $0 < \rho < 1$. For each edge xy , put $u = d_G(y)$ and $v = d_G(x)$, and let q, q' be the densities of its full and residual links, respectively. Set $q' = 0$ if a residual class is empty. These quantities depend on xy in every edge sum below.

The additive loss we allow is

$$\varepsilon = \frac{F_{k-1}}{k^2}, \quad (7.1)$$

and the corrected residual potential is

$$S_{xy} = \mathcal{P}_{k-1}(G'_{xy}) = H(u-1, v-1)(q')^{k+19}(1-q')^{20}. \quad (7.2)$$

In particular, $S_{xy} = 0$ if a residual class is empty. Our goal is

$$\frac{1}{e} \sum_{xy \in E(G)} S_{xy} \geq \mathcal{P}_k(G) - \varepsilon > F_{k-1}.$$

It suffices to prove the stronger logarithmic inequality

$$\frac{1}{e} \sum_{xy \in E(G)} \log(S_{xy} + \varepsilon) \geq \log \mathcal{P}_k(G). \quad (7.3)$$

The positive shift ε makes every logarithm defined, even when $S_{xy} = 0$. Indeed, once (7.3) is established, concavity of the logarithm, applied with equal weights $1/e$ to the e edges, gives

$$\log \mathcal{P}_k(G) \leq \frac{1}{e} \sum_{xy \in E(G)} \log(S_{xy} + \varepsilon) \leq \log \left(\frac{1}{e} \sum_{xy \in E(G)} S_{xy} + \varepsilon \right).$$

Here we used that ε is the same for every edge. Exponentiating and subtracting ε gives the required lower bound on the arithmetic average. The strict inequality follows from $\mathcal{P}_k(G) > F_k$ and $F_k - F_{k-1} > \varepsilon$. The inequalities will be proved differently in two different ranges of full-link densities: $q \leq (1+\rho)/2$, and $q > (1+\rho)/2$.

1. *The available estimates.* We use three previously established estimates. First, apply (4.7) with exponent $k+19 = (k-1) + 20$ and add $20 \log(1-\rho)$ to obtain

$$\frac{1}{e} \sum_{xy \in E(G)} \log(H(u, v)q^{k+19}(1-\rho)^{20}) \geq \log \mathcal{P}_k(G) + (k+19)\delta. \quad (7.4)$$

Second, Lemma 5.1 gives

$$H(u, v)q^{k+19} \leq H(u-1, v-1)(q')^{k+19} + \min\{2, 5(k+19)q^{k+18}\}.$$

Third, Lemma 6.1 gives

$$\begin{aligned} 0 &\leq \frac{1}{e} \sum_{xy \in E(G)} (q - \rho) \leq 10^4 \rho^{-5} \delta, \\ \frac{1}{e} \sum_{xy \in E(G)} (q - \rho)^2 &\leq 10^4 \rho^{-5} \delta. \end{aligned}$$

We will use the last two estimates to show that passing from the expression inside the logarithm in (7.4) to $S_{xy} + \varepsilon$ loses at most $k\delta$ on average.

Recall that $A = 2^{40}$ and $F_j \geq A$ for every $j \geq 1$. Now (3.20) implies

$$A \leq F_k < \mathcal{P}_k(G) = B_k(G)[\rho(1-\rho)]^{20} \leq 2(k-1)[\rho(1-\rho)]^{20}. \quad (7.5)$$

Consequently,

$$k > A/2, \quad [\rho(1-\rho)]^{20} > \frac{A}{2k}. \quad (7.6)$$

Thus, under the bound $B_k(G) \leq 2(k-1)$, the condition $\mathcal{P}_k(G) > F_k$ forces the parent density ρ away from both endpoints, by an amount depending on k . This restriction will control both the refined deletion loss and the coefficient of the averaged density error.

2. *Links with $q \leq (1+\rho)/2$.* We first bound the deletion error $5(k+19)q^{k+18}$ by ε . From (7.6), $(1-\rho)^{20} > A/(2k)$. Since $0 < 1-\rho < 1$, we obtain

$$1-\rho \geq \sqrt{\frac{A}{2k}}. \quad (7.7)$$

Using $q \leq 1 - (1-\rho)/2$, $1-s \leq \exp(-s)$, $F_{k-1} \geq A$, and $k > A/2 > 19$, we get

$$\frac{5(k+19)q^{k+18}}{F_{k-1}} \leq \frac{10k}{A} \exp\left(-\frac{k(1-\rho)}{2}\right) \leq \frac{10k}{A} \exp\left(-\sqrt{\frac{Ak}{8}}\right). \quad (7.8)$$

The inequality $\exp(x) \geq x^6/6!$ bounds the final expression by

$$\frac{10 \cdot 8^3 \cdot 6!}{A^4 k^2} = \frac{3,686,400}{A^4 k^2} \leq \frac{1}{k^2}.$$

Hence

$$5(k+19)q^{k+18} \leq \varepsilon. \quad (7.9)$$

Since $q' \leq q$, multiplying the deletion bound by $(1-q)^{20}$ gives

$$H(u, v)q^{k+19}(1-q)^{20} \leq S_{xy} + \varepsilon(1-q)^{20} \leq S_{xy} + \varepsilon. \quad (7.10)$$

Thus the factor $(1-q')^{20}$ causes no additional deletion loss. To compare with the parent factor, note that on the interval between ρ and q , we have $1-s \geq (1-\rho)/2$. The second derivative of $20 \log(1-s)$ is therefore at least $-80/(1-\rho)^2$, so Taylor's inequality gives

$$20 \log(1-q) \geq 20 \log(1-\rho) - \frac{20(q-\rho)}{1-\rho} - \frac{40(q-\rho)^2}{(1-\rho)^2}. \quad (7.11)$$

Combining this with the logarithm of (7.10) bounds the loss by a linear and a quadratic term in $q-\rho$.

3. *Links with $q > (1+\rho)/2$.* Here we use the constant deletion bound and the residual density hypothesis (3.21):

$$H(u, v)q^{k+19} \leq B_{k-1}(G'_{xy})(q')^{20} + 2 \leq 2(k-2) + 2 \leq 2k. \quad (7.12)$$

Since $S_{xy} + \varepsilon \geq \varepsilon \geq 1/k^2$, we have

$$\log \frac{H(u, v)q^{k+19}(1-\rho)^{20}}{S_{xy} + \varepsilon} \leq \log \frac{2k}{\varepsilon} \leq \log(2k^3) \leq 3 \log(2k).$$

The inequality $q-\rho > (1-\rho)/2$ now gives

$$3 \log(2k) \leq \frac{12 \log(2k)}{(1-\rho)^2} (q-\rho)^2.$$

The logarithmic loss is therefore bounded by a multiple of $(q-\rho)^2$, even when $q=1$ or $S_{xy}=0$.

4. *Averaging and completing the proof.* The estimates in the two ranges give, for every edge,

$$\begin{aligned} \log(S_{xy} + \varepsilon) &\geq \log(H(u, v)q^{k+19}(1-\rho)^{20}) \\ &\quad - \frac{20(q-\rho)}{1-\rho} - \frac{40+12\log(2k)}{(1-\rho)^2}(q-\rho)^2. \end{aligned} \quad (7.13)$$

In the first range, the extra quadratic term only weakens the bound from (7.10) and (7.11). In the second range, the additional terms subtracted on the right are nonnegative because $q - \rho > 0$. Having one inequality over all edges preserves the cancellation in the signed linear term when we average.

By Lemma 6.1, the average of the two error terms in (7.13) is at most

$$10^4 \rho^{-5} \left(\frac{20}{1-\rho} + \frac{40+12\log(2k)}{(1-\rho)^2} \right) \delta \leq \frac{10^4 \rho^{-5}}{(1-\rho)^2} (60+12\log(2k)) \delta. \quad (7.14)$$

To show that this is at most $k\delta$, use the density restriction (7.6):

$$\rho^{-5}(1-\rho)^{-2} \leq [\rho(1-\rho)]^{-5} < \left(\frac{2k}{A} \right)^{1/4}. \quad (7.15)$$

The coefficient of δ on the right of (7.14), divided by k , is therefore at most

$$10^4 \left(\frac{2}{A} \right)^{1/4} k^{-3/4} (60+12\log(2k)).$$

The function $x^{-3/4}(60+12\log(2x))$ decreases for $x \geq 1$, since its derivative has the sign of $12 - \frac{3}{4}(60+12\log(2x)) < 0$. As $k > A/2$, the last display is at most

$$\frac{2 \cdot 10^4}{A} (60+12\log A) < 1. \quad (7.16)$$

Indeed, $\log A < 28$ and $A = 2^{40} > 10^{12}$, whereas the numerator is less than 7,920,000.

Averaging (7.13) consequently gives

$$\frac{1}{e} \sum_{xy \in E(G)} \log(S_{xy} + \varepsilon) \geq \frac{1}{e} \sum_{xy \in E(G)} \log(H(u, v)q^{k+19}(1-\rho)^{20}) - k\delta. \quad (7.17)$$

Combining this with (7.4) and $\delta \geq 0$ yields

$$\frac{1}{e} \sum_{xy \in E(G)} \log(S_{xy} + \varepsilon) \geq \log \mathcal{P}_k(G) + 19\delta \geq \log \mathcal{P}_k(G), \quad (7.18)$$

which proves (7.3) and, by the reduction at the beginning, completes the proof. $\square$

7.2 Deducing a biclique from the potential

We next combine Theorem 3.1 with the preliminary density criterion to obtain a statement for arbitrary bipartite graphs.

Corollary 7.1. *For every positive integer k and every bipartite graph G ,*

$$\mathcal{P}_k(G) > F_k \implies G \text{ contains } K_{k,k}. \quad (7.19)$$

Proof. We induct on k . For $k = 1$, positive potential implies positive density, so G has an edge. Now let $k \geq 2$, assume the result for $k - 1$, and suppose $\mathcal{P}_k(G) > F_k$.

If $B_k(G) > 2(k - 1)$, then [Lemma 5.2](#) gives a $K_{k,k}$ in G directly. We may therefore assume $B_k(G) \leq 2(k - 1)$.

If some edge xy has $B_{k-1}(G'_{xy}) > 2(k - 2)$, then [Lemma 5.2](#), applied with $k - 1$ to that residual link, gives a $K_{k-1,k-1}$. Adding x, y yields a $K_{k,k}$ in G . We may therefore also assume $B_{k-1}(G'_{xy}) \leq 2(k - 2)$ for every edge.

The hypotheses of [Theorem 3.1](#) now hold, so some edge satisfies $\mathcal{P}_{k-1}(G'_{xy}) > F_{k-1}$. The induction hypothesis gives a $K_{k-1,k-1}$ in that residual link, which again extends to a $K_{k,k}$ in G . $\square$

7.3 Substitution of $N = c2^t$

We now deduce [Theorem 2.1](#) and [Corollary 2.2](#) from [Corollary 7.1](#). For a graph with two classes of size N and density exactly $1/2$, the potential is

$$\mathcal{P}_t(G) = N2^{-t-40}.$$

Thus $N \geq 2^{t+81}$ makes the potential at least $2^{41} > F_t$. The following reduction handles starting density at least $1/2$.

Proof of Theorem 2.1. Put $M = 2^{t+81}$. If $N \geq M$ and $e(G) \geq N^2/2$, sum $e_G(S, T)$ over all M -element sets $S \subseteq L, T \subseteq R$. Every edge is counted $\binom{N-1}{M-1}^2$ times, so the mean is

$$\frac{e(G)\binom{N-1}{M-1}^2}{\binom{N}{M}^2} = e(G)(M/N)^2 \geq M^2/2.$$

Some pair S, T therefore spans at least $M^2/2$ edges. Since M is even, keep exactly $M^2/2$ of those edges. The resulting graph J has two classes of size M and density exactly $1/2$, so

$$\mathcal{P}_t(J) = M2^{-t-40} = 2^{41} > F_t.$$

By [Corollary 7.1](#), J contains a $K_{t,t}$, and hence so does G . $\square$

The passage to exactly half the edges is necessary because the potential in (3.17) is not increasing throughout $0 < \rho < 1$. We do not simply substitute $\rho = 1/2$ into an inequality whose actual density might be larger.

Proof of Corollary 2.2. In a red–blue coloring of $K_{N,N}$, one color has at least $N^2/2$ edges. For $N = 2^{t+81}$, [Theorem 2.1](#) applied to that color gives a monochromatic $K_{t,t}$. $\square$

A Standard inequalities

The purpose of this appendix is to record and prove the two elementary convexity inequalities used in the main text. All graph-specific lemmas and their proofs are in the sections where they are needed.

Lemma A.1. *If $\lambda_1, \dots, \lambda_s \geq 0$ have sum one and $x_1, \dots, x_s > 0$, then*

$$\sum_{a=1}^s \lambda_a \log x_a \leq \log \left(\sum_{a=1}^s \lambda_a x_a \right). \quad (\text{A.1})$$

Proof. Put $M = \sum_a \lambda_a x_a$. The elementary inequality $\log z \leq z - 1$, applied to $z = x_a/M$ and multiplied by λ_a , gives

$$\sum_a \lambda_a \log(x_a/M) \leq \sum_a \lambda_a (x_a/M - 1) = 0.$$

This is (A.1). □

Lemma A.2. *For $s \geq 0$ and $t > 0$, put*

$$\psi(s, t) = s \log(s/t) - s + t, \tag{A.2}$$

where $0 \log(0/t) = 0$. Then $\psi(s, t) \geq 0$. If $0 \leq s \leq M$ and $0 < t \leq M$, then

$$\psi(s, t) \geq \frac{(s - t)^2}{2M}. \tag{A.3}$$

We also put $\psi(0, 0) = 0$.

Proof. For $F(x) = x \log x$, the expression in (A.2) is

$$F(s) - F(t) - F'(t)(s - t).$$

Since $F''(x) = 1/x > 0$ for $x > 0$, this expression is nonnegative. On $(0, M]$, we have $F''(x) \geq 1/M$, so Taylor's inequality gives (A.3). The statements at $s = 0$ follow by continuity. □

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